

Deepthika S

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Available: Immediate | Open to Bangalore, Chennai, Hyderabad

Summary

Software Engineer with 3+ years at Citigroup and Morgan Stanley in large-scale fintech systems, specialising in AI/ML engineering. Built and shipped LLM pipelines, multi-agent systems (LangChain, LangGraph), and RAG architectures independently as personal and open-source projects. Open-source contributor to IBM's AI Alliance. M.S. Computer Science (AI/ML), GPA 3.96/4.0, SUNY Buffalo (Dec 2025). Core stack: Python, Java, PyTorch, HuggingFace, FastAPI.

Technical Skills

GenAI & LLMs:	LangChain, LangGraph, RAG, Embeddings, Fine-tuning, OpenAI API, Gemini API, LlamaIndex, Prompt Engineering
ML & Deep Learning:	PyTorch, TensorFlow, HuggingFace, Scikit-learn, ONNX, YOLOv8, ResNet-50, EfficientNet-B0
Languages:	Python, Java, SQL, JavaScript, TypeScript, C#
Backend & Infra:	FastAPI, Flask, Spring Boot, Docker, Kubernetes, Jenkins, GitHub Actions, Firebase, Vercel
Databases:	MySQL, PostgreSQL, MongoDB, Firestore, ChromaDB, Vector Databases
Domain:	Fintech, Investment Banking, Wealth Management, APAC Financial Markets

AI/ML Projects

Event Connect - AI Attendee-Sponsor Matcher - *LangChain, LangGraph, OpenRouter, Vercel, Firestore* **2026**
Open Source Contributor: AI Alliance (IBM-led initiative) github.com/deepthika-sivaram/event-matcher

- Decomposed a monolithic LLM call into a 5-stage **LangGraph agentic pipeline**, reducing LLM API calls 60–80% via embedding-based matching and multi-layer Firestore **caching**, cutting repeat-run latency from 12s to under 1s.
- Deployed on **Vercel serverless** with fan-out parallel execution across agents and local agent **fallback** ensuring zero downtime; **parallelized** narrative generation cutting stage latency 57% (11,188ms to 4,803ms).

Engineering Knowledge Agent - Multi-Source RAG System - *ChromaDB, Gemini, LlamaIndex, Streamlit* **2026**
Personal Project github.com/deepthika-sivaram/engineering-knowledge-agent

- Built RAG pipeline ingesting GitHub PRs, commits and Confluence-style docs into **ChromaDB** with **per-source typed metadata** - enabling **query-time source filtering** and **per-source inline citations** in a Streamlit UI.
- Extended with **LlamaIndex connectors** pulling live GitHub code, PRs, issues and commit history via REST API; strict context-only prompting to **prevent hallucination**.

AI Sketch-to-UI Converter - *PyTorch, YOLOv8, CNN, Self-Attention, Transformer, GRU, EasyOCR* **2025**
Academic Project: SUNY Buffalo github.com/deepthika-sivaram/AI-Sketch-to-UI-Converter

- Fine-tuned** YOLOv8s across 14 UI classes achieving mAP@50: 0.957 — built CNN→Attention→Transformer→GRU pipeline with EasyOCR generating position-accurate multi-themed HTML from hand-drawn wireframes end-to-end.

Experience

Software Engineer **Jul 2021 - Jul 2024**
Citicorp Services India Private Limited *Chennai, India*

- Spearheaded **GitHub Copilot adoption** across 7 engineers including non-technical testers - trained testers on structured Gherkin prompting to independently generate automation scripts, improving team productivity **~40%**.
- Built the **Java/Spring Boot** engineering framework from scratch (Selenium, Cucumber, RestAssured) scaling regression coverage from **15% to 85%** across 500+ endpoints, compressing 3-day manual cycles to **under 12 hours**.
- Built automated PDF data extraction and cross-validation system (**Apache PDFBox**) for compliance-critical financial statements, reducing manual validation time by **~80%** across 20+ document validations per release.
- Led **Citi's Southeast Asia divestiture** as Wealth workstream lead - built reusable order lifecycle automation across 5 countries, owned a **critical settlement failure**, and delivered **zero data loss** within **5-7 days** with no playbook.
- Engineered **cross-platform mobile automation** (Selenium, Perfecto) for Citi's wealth management platform serving **millions of APAC users** - adopted by other product teams as the **shared framework foundation**.

Software Development Intern **May 2020 - Jul 2020**
Morgan Stanley *Bangalore, India*

- Containerised a Spring Boot **POC application** with Docker and Kubernetes, deploying to Morgan Stanley's **private cloud** via a Jenkins **CI/CD pipeline** and **cutting deployment time 75% (from 40 min to under 10 min)**.
- Delivered the POC as a **reusable cloud-native template** adopted by the entire team for **12-factor compliant** service development; integrated **OAuth 2.0/OIDC authentication** via Spring Security.

Education

M.S. Computer Science Engineering (AI/ML) - *State University of New York at Buffalo* **GPA: 3.96/4** Dec 2025

B.E. Computer Science Engineering - *College of Engineering Guindy (Anna University)* **GPA: 9.26/10** May 2021

Recognition & Leadership

- Citi Gratitude Awards (Bronze & Copper, 2023 & 2024)** - Top 5% of engineers for technical innovation.
- Designated **SME, Fintech & Engineering** - Mentored 6 engineers on engineering standards and delivery practices.